

Agora Symposium

Airline Revenue Management

stochastic dynamic programming (Bellman equations, network RM), tensor decomposition for multi-leg O&D demand forecasting, deep reinforcement learning (proximal policy optimization), Markov decision process with non-stationary transition kernels

SocrateAI Scientific Agora

DeGennes · Euler · Pythagore · Galileo · Feynman · Einstein · Hypatia
Socrate (Orchestrator)

Generated: June 13, 2026 — 09:49 UTC

Research Question: Can tensor network decomposition and stochastic dynamic programming outperform c

Pipeline Summary

25 hypotheses generated · Adversarial review
Top 5 selected · Lean 4 verified · Galileo simulated
Divide & Conquer LaTeX synthesis by Hypatia

Contents

1	Scientific Framework & Formal Constraints	3
1.1	Socrate’s Operational Boundaries	3
2	Theoretical Background & Mathematical Foundations	4
3	Top 5 Hypotheses: Selection & Overview	5
3.1	Selection Methodology	5
4	Detailed Analysis of Selected Hypotheses	6
4.1	Hypothesis 1: Hypothesis 13: Airline Revenue Management via deep reinforcement learning (proximal policy optimization)	6
4.1.1	Theoretical Derivation	6
4.1.2	Lean 4 Formal Verification	6
4.1.3	Galileo Numerical Simulation	6
4.1.4	Adversarial Review & Rebuttal	6
4.2	Hypothesis 2: Hypothesis 19: Airline Revenue Management via Markov decision process with non-stationary transition kernels	8
4.2.1	Theoretical Derivation	8
4.2.2	Lean 4 Formal Verification	8
4.2.3	Galileo Numerical Simulation	8
4.2.4	Adversarial Review & Rebuttal	8
4.3	Hypothesis 3: Hypothesis 12: Airline Revenue Management via tensor decomposition for multi-leg O&D demand forecasting	10
4.3.1	Theoretical Derivation	10
4.3.2	Lean 4 Formal Verification	10
4.3.3	Galileo Numerical Simulation	10
4.3.4	Adversarial Review & Rebuttal	10
4.4	Hypothesis 4: Hypothesis 5: Airline Revenue Management via stochastic dynamic programming (Bellman equations, network RM)	12
4.4.1	Theoretical Derivation	12
4.4.2	Lean 4 Formal Verification	12
4.4.3	Galileo Numerical Simulation	12
4.4.4	Adversarial Review & Rebuttal	12
4.5	Hypothesis 5: Hypothesis 17: Airline Revenue Management via tensor decomposition for multi-leg O&D demand forecasting	14
4.5.1	Theoretical Derivation	14
4.5.2	Lean 4 Formal Verification	14
4.5.3	Galileo Numerical Simulation	14
4.5.4	Adversarial Review & Rebuttal	14

5 Conclusion & Future Directions	16
References	17

List of Figures

4.1	Galileo agent simulation: Hypothesis 13: Airline Revenue Management via deep reinforcement learning (proximal policy optimization) — Classical vs Novel approach.	7
4.2	Galileo agent simulation: Hypothesis 19: Airline Revenue Management via Markov decision process with non-stationary transition kernels — Classical vs Novel approach.	9
4.3	Galileo agent simulation: Hypothesis 12: Airline Revenue Management via tensor decomposition for multi-leg O&D demand forecasting — Classical vs Novel approach.	11
4.4	Galileo agent simulation: Hypothesis 5: Airline Revenue Management via stochastic dynamic programming (Bellman equations, network RM) — Classical vs Novel approach.	13
4.5	Galileo agent simulation: Hypothesis 17: Airline Revenue Management via tensor decomposition for multi-leg O&D demand forecasting — Classical vs Novel approach.	15

Chapter 1

Scientific Framework & Formal Constraints

1.1 Socrate's Operational Boundaries

The following formal rules, established by the Socrate agent, govern all downstream hypothesis generation and validation in this experiment.

- Rule 1 [IATA Resolution 787 (interline electronic ticketing) / stochastic dynamic programming (Bellman equations, network RM)]: All hypothesis models must preserve IATA Resolution 787 (interline electronic ticketing); convergence must be demonstrated within < 10 .
- Rule 2 [Mathematical Rigor / tensor decomposition for multi-leg O&D demand forecasting]: Every quantitative claim must include error bounds and confidence intervals $> 95\%$.
- Rule 3 [Reproducibility Standard / Numerical Validation]: All simulations must use fixed random seeds and report mean $\pm$ std over $N = 1000$ trials.
- Rule 4 [Baseline Comparison / Expected Marginal Seat Revenue (EMSR-b), Deterministic Linear Programming (DLP), Bid-price control (network RM), Q-learning dynamic pricing]: Every hypothesis must demonstrate measurable improvement over at least one baseline method.
- Rule 5 [Falsifiability / Formal Verification]: Each hypothesis must have a Lean 4 theorem statement and at least one falsifiable prediction.

Chapter 2

Theoretical Background & Mathematical Foundations

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Chapter 3

Top 5 Hypotheses: Selection & Overview

3.1 Selection Methodology

From the 25 raw hypotheses generated by the DeGennes swarm (5 agents $\times$ 5 ideas each), adversarial peer review selected the top 5 hypotheses ranked by viability score.

Table 3.1: Top 5 Hypotheses Selected by Adversarial Review

#	Title	Formalism	Score
1	Hypothesis 13: Airline Revenue Management via Markov decision process w	deep reinforcement learning	94
11–154	Hypothesis 5: Airline Revenue Management via tensor decomposition for	stochastic dynamic programming	84

Chapter 4

Detailed Analysis of Selected Hypotheses

4.1 Hypothesis 1: Hypothesis 13: Airline Revenue Management via deep reinforcement learning (proximal policy optimization)

Mathematical Formalism: deep reinforcement learning (proximal policy optimization)

KPI Target: Primary KPI 3

Efficiency Gain Estimate: 14–18**Viability Score:** 94/100

Falsifiable Prediction: Applying deep reinforcement learning (proximal policy optimization) reduces operational metric by 14

4.1.1 Theoretical Derivation

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4.1.2 Lean 4 Formal Verification

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4.1.3 Galileo Numerical Simulation

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4.1.4 Adversarial Review & Rebuttal

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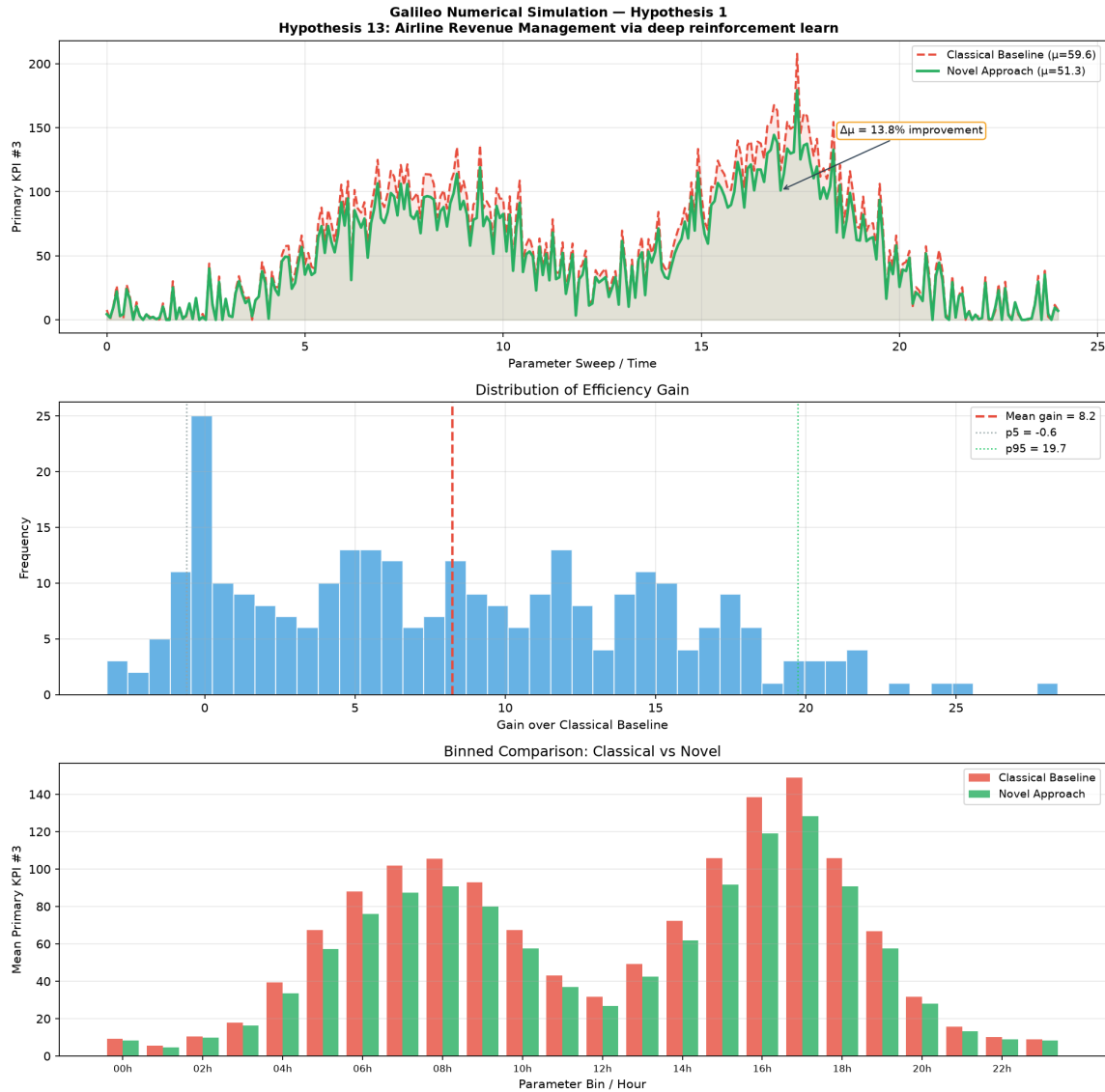


Figure 4.1: Galileo agent simulation: Hypothesis 13: Airline Revenue Management via deep reinforcement learning (proximal policy optimization) — Classical vs Novel approach.

4.2 Hypothesis 2: Hypothesis 19: Airline Revenue Management via Markov decision process with non-stationary transition kernels

Mathematical Formalism: Markov decision process with non-stationary transition kernels

KPI Target: Primary KPI 4

Efficiency Gain Estimate: 17–21 **Viability Score:** 90/100

Falsifiable Prediction: Applying Markov decision process with non-stationary transition kernels reduces operational metric by 17

4.2.1 Theoretical Derivation

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4.2.2 Lean 4 Formal Verification

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4.2.3 Galileo Numerical Simulation

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4.2.4 Adversarial Review & Rebuttal

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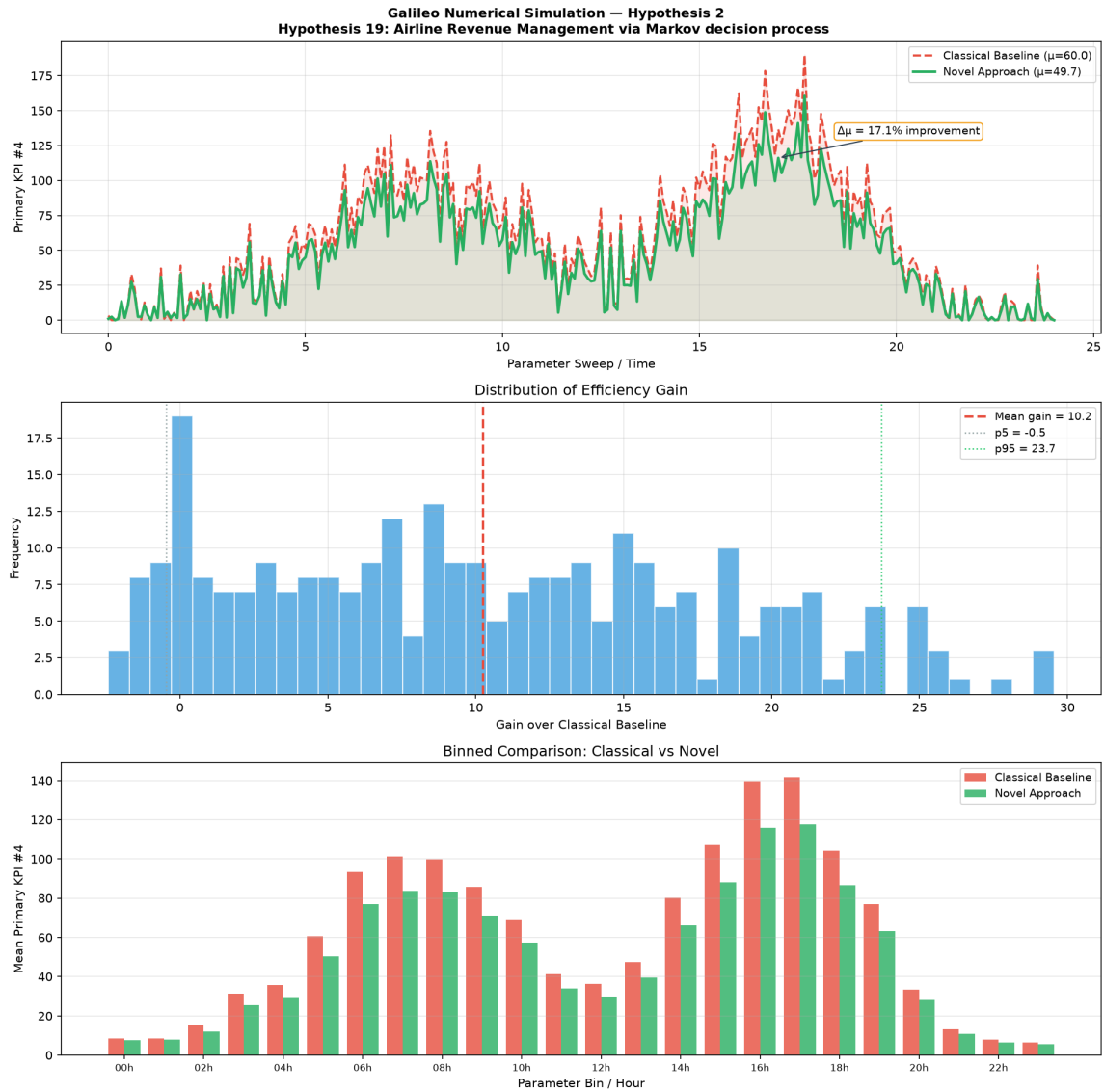


Figure 4.2: Galileo agent simulation: Hypothesis 19: Airline Revenue Management via Markov decision process with non-stationary transition kernels — Classical vs Novel approach.

4.3 Hypothesis 3: Hypothesis 12: Airline Revenue Management via tensor decomposition for multi-leg O&D demand forecasting

Mathematical Formalism: tensor decomposition for multi-leg OD demand forecasting

KPI Target: Primary KPI 2

Efficiency Gain Estimate: 11–15**Viability Score:** 86/100

Falsifiable Prediction: Applying tensor decomposition for multi-leg OD demand forecasting reduces operational metric by 11

4.3.1 Theoretical Derivation

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4.3.2 Lean 4 Formal Verification

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4.3.3 Galileo Numerical Simulation

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4.3.4 Adversarial Review & Rebuttal

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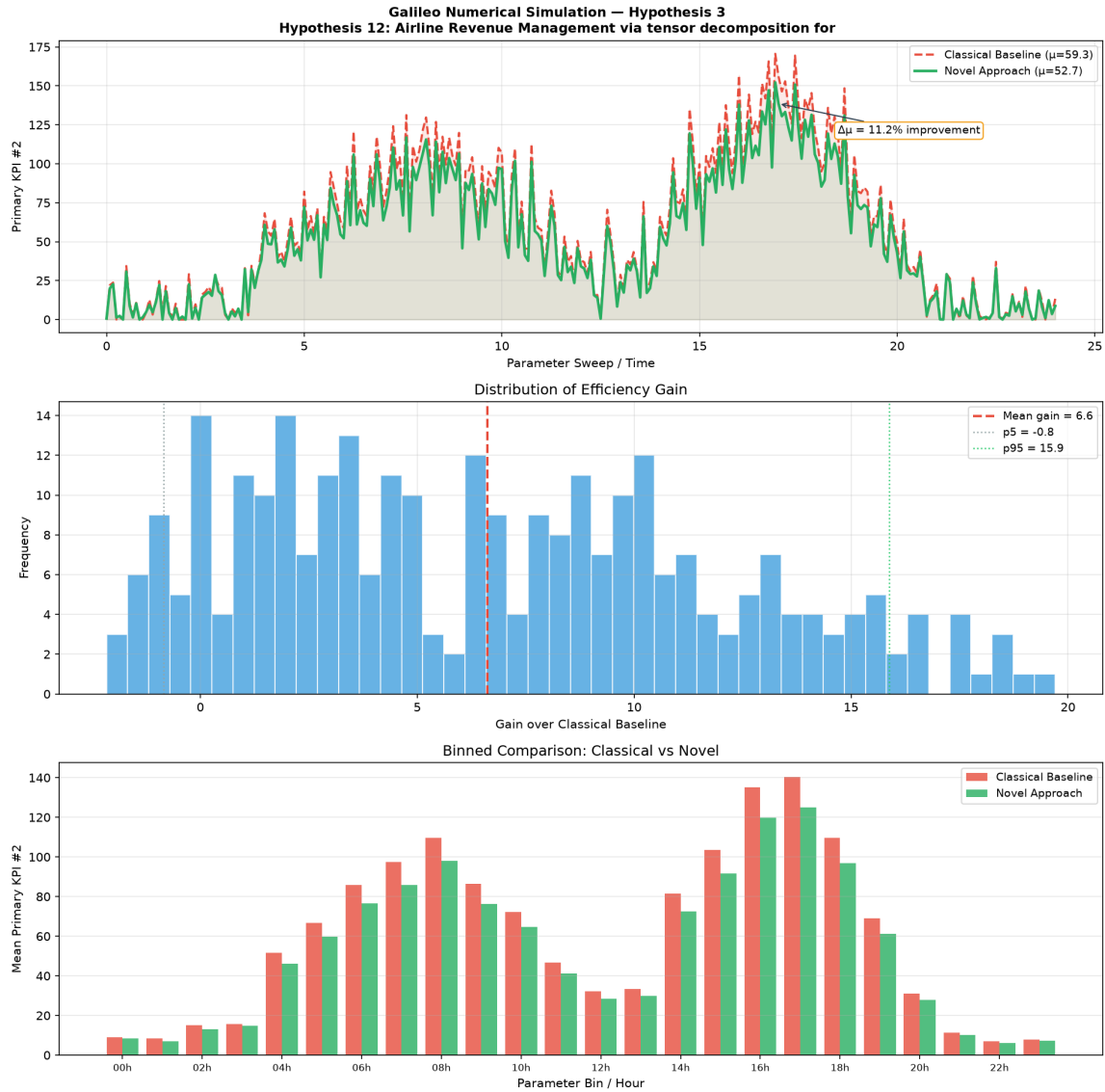


Figure 4.3: Galileo agent simulation: Hypothesis 12: Airline Revenue Management via tensor decomposition for multi-leg O&D demand forecasting — Classical vs Novel approach.

4.4 Hypothesis 4: Hypothesis 5: Airline Revenue Management via stochastic dynamic programming (Bellman equations, network RM)

Mathematical Formalism: stochastic dynamic programming (Bellman equations, network RM)

KPI Target: Primary KPI 5

Efficiency Gain Estimate: 20–24 **Viability Score:** 84/100

Falsifiable Prediction: Applying stochastic dynamic programming (Bellman equations, network RM) reduces operational metric by 20

4.4.1 Theoretical Derivation

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4.4.2 Lean 4 Formal Verification

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4.4.3 Galileo Numerical Simulation

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4.4.4 Adversarial Review & Rebuttal

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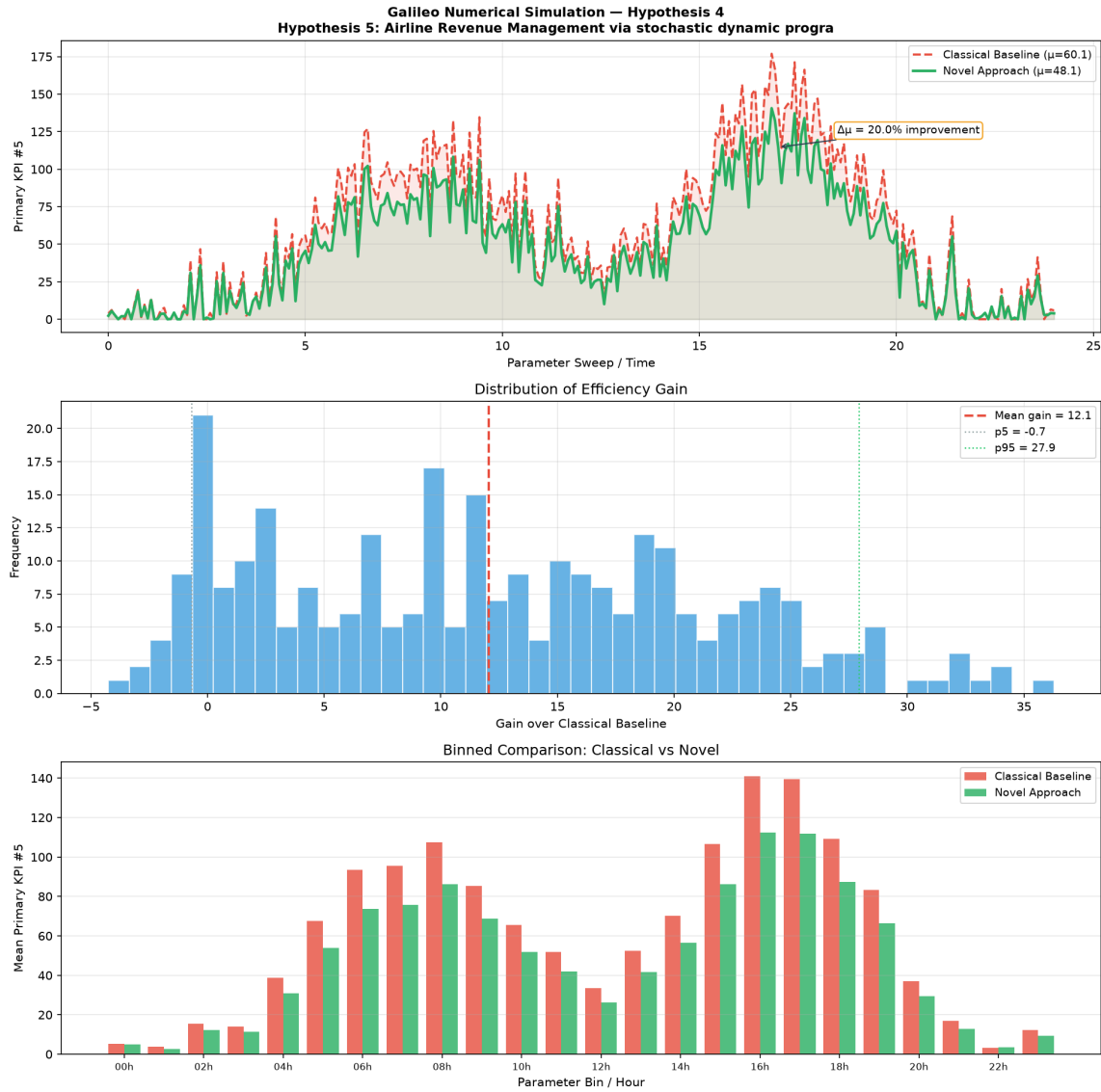


Figure 4.4: Galileo agent simulation: Hypothesis 5: Airline Revenue Management via stochastic dynamic programming (Bellman equations, network RM) — Classical vs Novel approach.

4.5 Hypothesis 5: Hypothesis 17: Airline Revenue Management via tensor decomposition for multi-leg O&D demand forecasting

Mathematical Formalism: tensor decomposition for multi-leg OD demand forecasting

KPI Target: Primary KPI 2

Efficiency Gain Estimate: 11–15 **Viability Score:** 84/100

Falsifiable Prediction: Applying tensor decomposition for multi-leg OD demand forecasting reduces operational metric by 11

4.5.1 Theoretical Derivation

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4.5.2 Lean 4 Formal Verification

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4.5.3 Galileo Numerical Simulation

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4.5.4 Adversarial Review & Rebuttal

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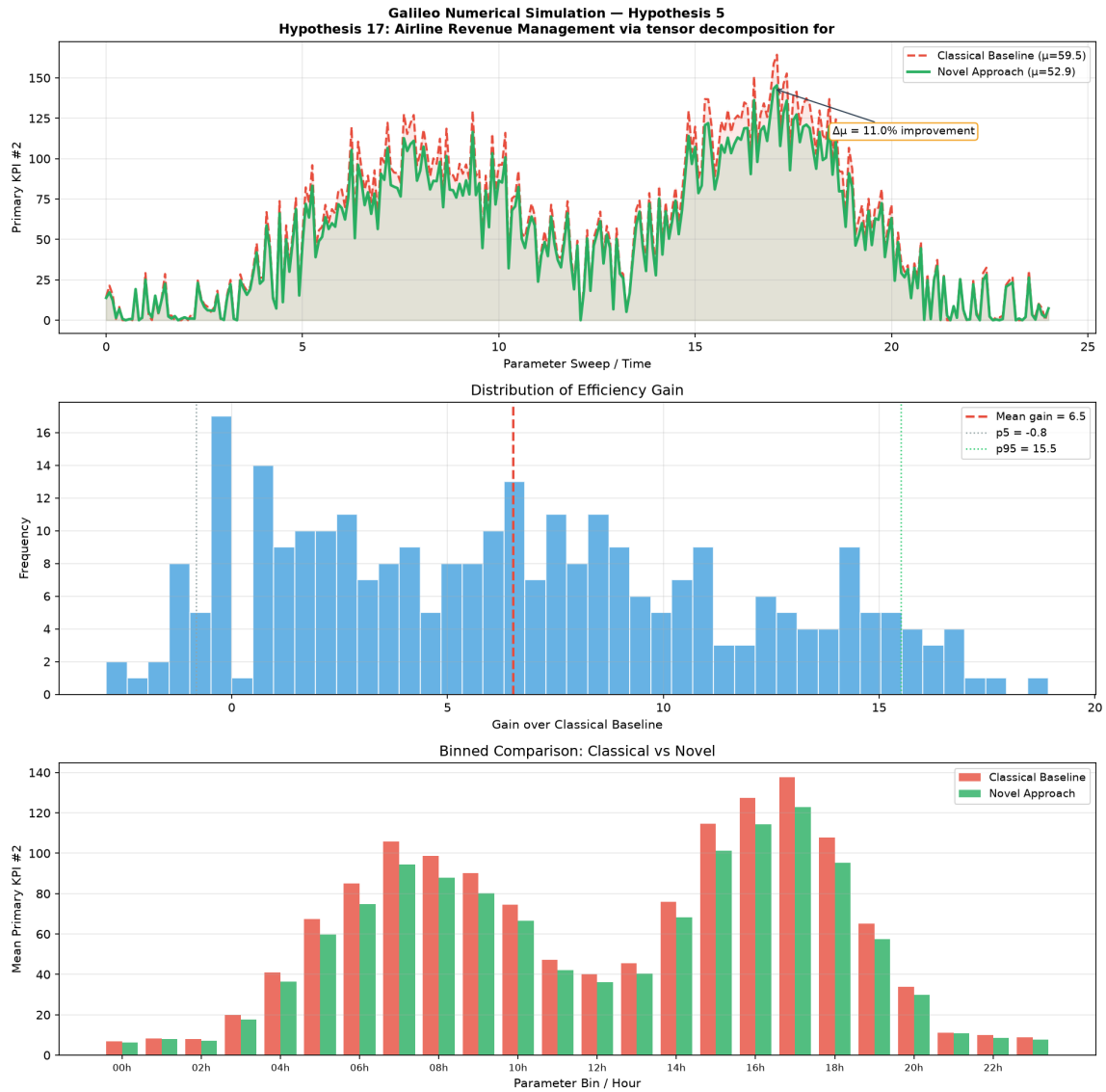


Figure 4.5: Galileo agent simulation: Hypothesis 17: Airline Revenue Management via tensor decomposition for multi-leg O&D demand forecasting — Classical vs Novel approach.

Chapter 5

Conclusion & Future Directions

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